



United International University
School of Science and Engineering
Final Examination Trimester: Summer 2024
Course Title: Coordinate Geometry and Vector Analysis
Course Code: Math 2201 Marks: 50
Total Time: 2 hours

Answer all questions.

1. a) Consider, $F(x, y) = (2xy + y^2 + 1)i + (x^2 + 2xy + 1)j$ [7]
 - i) Show that F is a conservative vector field on the entire xy -plane.
 - ii) Find the potential function $\phi(x, y)$.
 - iii) Find $\int_{(0,0)}^{(2,3)} F \cdot dr$ using ii)b) Using Green's theorem find the value of $\oint_C F \cdot dr$
Where $F(x, y) = (\sin x e^{3x} - 5y^2)i + (5y^3 + 2x)j$ and C is the closed curve with parametric equations $x = 3\cos t$, and $y = 3\sin t$. [6]
2. a) Evaluate $\int_C (x + y)dx + (-y)dy$ along the triangle with vertices $(0, 0)$, $(0, 1)$ and $(1, 0)$. [6]
b) Evaluate the surface integral $\iint_{\sigma} yz \, ds$; σ is the part of the plane $2x + y + z = 2$ that lies in the first octant. [6]
3. a) Find the flux of the vector field $F(x, y, z) = x i + y j + 2z k$ across σ , [7]
where σ is the portion of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 2$ and $z = 5$, oriented upward unit normal.
b) Use the Divergence Theorem to find the outward flux of the vector field $F(x, y, z) = x^3 i + y^3 j + z^3 k$ across the surface of the region that is enclosed by $z = 36 - x^2 - y^2$ and the plane $z = 0$. [6]
4. a) Using triple integral find the volume of the solid bounded by the $x^2 + y^2 + z^2 = 4$ and xy -plane. [6]
b) Identify and Sketch the Conic $4x^2 - 25y^2 - 16x + 150y - 125 = 0$. [6]